NOT MEASUREMENT SENSITIVE

MIL-P-85856/13
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS
NICKEL BASE ALLOY, TYPE I, CLASS 2, COMPOSITION K

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Mo	Cr	Ni	Fe	V	C	Mn	Si	Co	P	S
23.0 26.0	4.0 6.0	Bal	4.0 7.0	0.6 max.	0.12 max.	1.0 max.	1.0 max.	2.5 max.	0.02 max.	0.02 max.

Particle Size/Distribution

+120 Mesh (+125 µm)	1.0 max.
+140 Mesh (+106 µm)	10.0 max.
+325 Mesh (+45 µm)	80.0 min.

Sprayed Coating Properties

Bond Strength	31.0 MPa (4500 psi) min.
Hardness	Rockwell 15T, 90 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

AMSC N/A FSC 3439
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Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 9.0 lb/hr
35% min for A/H_2 @ 8.5 lb/hr

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/13)

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